

SnS Homework III

Problem 1 [12 pts]

Solve the following difference equation using z-transforms:

$$y[n] + 3y[n-1] + 2y[n-2] = 2x[n] - x[n-1]; y[-1] = 0; y[-2] = 1, x[n] = u[n]$$

$$\therefore \mathcal{Z}\{y[n-k]\} = z^{-k} Y[z] + z^{-k+1} y[-1] \dots z^{-1} y[-k+1] \\ \text{for } k > 0$$

• $k=1$:

$$\mathcal{Z}\{y[n-1]\} = z^{-1} Y[z] + y[-1] \\ = z^{-1} Y[z]$$

• $k=2$:

$$\mathcal{Z}\{y[n-2]\} = z^{-1} Y[z] + y[-2] + 0 \\ = z^{-2} Y[z] + 1$$

• So, L.H.S becomes:

$$= Y[z] + 3[z^{-1} Y[z]] + 2[z^{-2} Y[z] + 1]$$

$$= Y[z] + 3 z^{-1} Y[z] + 2 z^{-2} Y[z] + 2$$

• Now, for R.H.S:-

$$x[n] = u[n]$$

$$X[z] = \frac{z}{z-1}$$

$$\mathcal{Z}\{x[n-1]\} = \left(\frac{1}{z}\right) \frac{z}{z-1} = \frac{1}{z-1}$$

So now, R.H.S becomes:

$$= \frac{2z}{z-1} - \left(\frac{1}{z-1}\right) = \frac{2z-1}{z-1}$$

Putting values in the equation:

$$Y[z] + 3 z^{-1} Y[z] + 2 z^{-2} Y[z] + 2 = \frac{2z-1}{z-1}$$

$$X[z] + 3z^{-1} Y[z] + 2z^{-2} Y[z] + 2 = \frac{2z-1}{z-1}$$

$$Y[z] (1 + 3z^{-1} + 2z^{-2}) + 2 = \frac{2z-1}{z-1}$$

$$Y[z] \left(1 + \frac{3}{z} + \frac{2}{z^2} \right) + 2 = \frac{2z-1}{z-1}$$

$$Y[z] \left(\frac{z^2 + 3z + 2}{z^2} \right) = \frac{2z-1}{z-1} - 2$$

$$Y[z] (z+2)(z+1) = \frac{z (2z-1-2z+2)}{z-1}$$

$$Y[z] = \frac{z^2}{(z+1)(z+2)(z-1)}$$

• solve using partial fractions:

$$\frac{Y[z]}{z} = \frac{z}{(z+1)(z+2)(z-1)}$$

Let:-

$$\frac{z}{(z+1)(z+2)(z-1)} = \frac{A}{(z+1)} + \frac{B}{(z+2)} + \frac{C}{(z-1)} \quad \text{--- i}$$

$$z = A(z+2)(z-1) + B(z+1)(z-1) + C(z+2)(z+1)$$

for $z = -1$:

$$-1 = A(1)(-2)$$

$$\boxed{A = \frac{1}{2}}$$

for $z = -2$:

$$-2 = B(-1)(-3)$$

$$\boxed{B = -2/3}$$

for $z = 1$:

$$1 = C(3)(2)$$

$$\boxed{C = 1/6}$$

Now, putting values of A, B, C in equation(i)

$$= \frac{1}{2} \cdot \frac{1}{(z+1)} + \left(-\frac{2}{3}\right) \left(\frac{1}{z+2}\right) + \left(\frac{1}{6}\right) \left(\frac{1}{z-1}\right)$$

$$\therefore z^{-1} \left\{ \frac{z}{z-a} \right\} = a^n u[n]$$

So,

$$y[n] = \frac{1}{2} (-1)^n u[n] - \frac{2}{3} u[n] + \frac{1}{6} u[n]$$

$$y[n] = u[n] \left(\frac{(-1)^n}{2} - \frac{2}{3} \cdot (-2^n) + \frac{1}{6} \right) \text{ Ans.}$$

Problem 2 [12 pts]

Find the z transform of the following signals:

Also, plot the ROC in the Z-domain for each part.

a) $x[n] = u[n] - u[n-4]$

b) $x[n] = 0.5^n u[n]$

c) $x[n] = \begin{matrix} \downarrow \\ [1 \ 4 \ 8 \ 2] \end{matrix}$

a.

$$\therefore z \{ u[n] \} = \frac{z}{z-1}$$

$$z \{ u[n-4] \} = z^{-4} z \{ u[n] \}$$

$$= \frac{z^{-3}}{z-1}$$

$$X[z] = \frac{z}{z-1} - \frac{z^{-3}}{z-1}$$

$$= \frac{z - z^{-3}}{(z-1)}$$

$$\frac{1}{(z-1)}$$

Solving:

$$= \frac{z^4 - 1}{z^3(z-1)} = \frac{(z^2)^2 - 1}{z^3(z-1)}$$

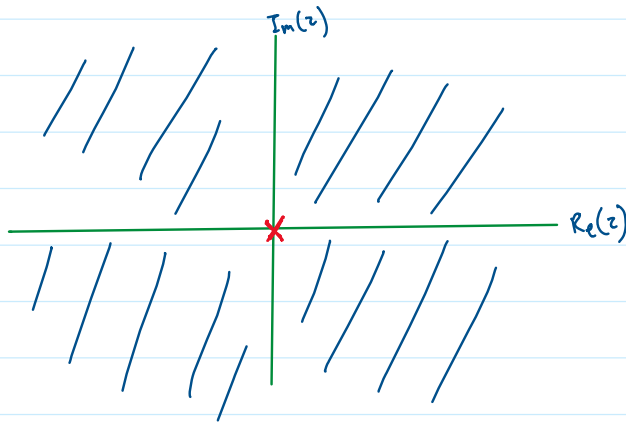
$$= \frac{(z^2-1)(z^2+1)}{z^3(z-1)} = \frac{\cancel{(z-1)}(z+1)(z^2+1)}{z^3\cancel{(z-1)}}$$

$$= \frac{(z+1)(z^2+1)}{z^3}$$

$$= \frac{z^3 + z^2 + z + 1}{z^3}$$

$$= 1 + \frac{1}{z} + \frac{1}{z^2} + \frac{1}{z^3}$$

as we see $z \neq 0$ so, $\text{ROC} = (z|>0)$



b)

$$\therefore a^n u[n] \rightarrow \frac{z}{z-a}, \quad |z| > |a|$$

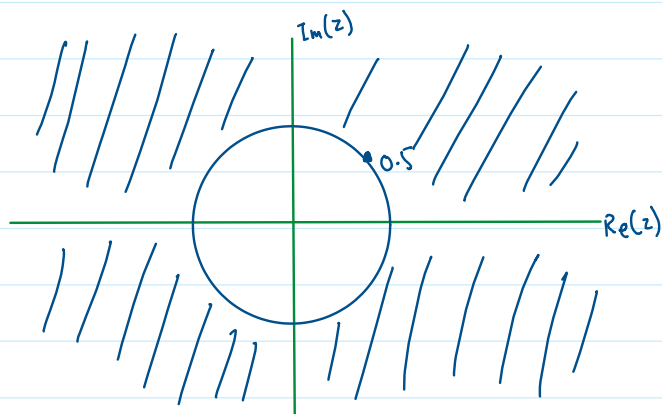
Hence,

$$X[z] = \frac{z}{z-0.5}$$

So,

$$z \neq 0.5$$

$$\text{ROC} = |z| > 0.5$$



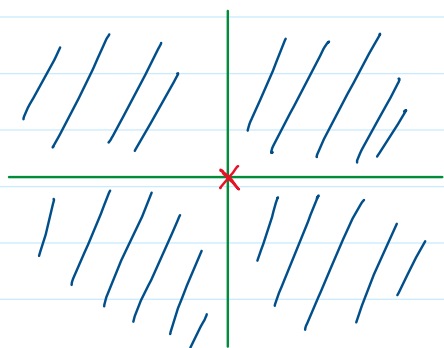
c) $x[n] = [1 \ 4 \ 8 \ 2]$

$$\therefore X[z] = \sum_{k=-\infty}^{\infty} x[k] z^{-k}$$

$$X[z] = 1 + 4z^{-1} + 8z^{-2} + 2z^{-3}$$

as $z \neq 0$

• for ROC: $ROC = |z| > 0$



Problem 3 [12 pts]

Using only the fact that $\gamma^n u[n] \longleftrightarrow z/(z - \gamma)$ and properties of the z -transform, find the z -transform of each of the following:

(a) $n^2 u[n]$

(b) $n^2 \gamma^n u[n]$

(c) $n^3 u[n]$

(d) $a^n [u[n] - u[n - m]]$

a) $\therefore n x[n] \xrightarrow{z} -z \frac{d}{dz} x[z]$

$$= n[n u[n]]$$

So,

After applying z-transform:

$$= n \left[-z \frac{d}{dz} \left(\frac{z}{z-1} \right) \right]$$

$$= n \left[-z \times \frac{\cancel{z}-1-\cancel{z}}{(z-1)^2} \right]$$

$$= n \frac{z}{(z-1)^2}$$

applying z-transform:

$$= -z \frac{d}{dz} \left(\frac{z}{(z-1)^2} \right)$$

$$= \frac{-z [(z-1)^2 - 2z(z-1)]}{(z-1)^4}$$

$$= \frac{z(-z-1)}{(z-1)^3} = \boxed{\frac{z(z+1)}{(z-1)^3}} \text{ Ans}$$

b) $n^2 r^n u[n]$

$$= n^2 [r^n u[n]] \quad \left[\because a^n u[n] \xrightarrow{z} \frac{z}{z-a} \right]$$

$$= n^2 \frac{z}{z-r}$$

$$= n \left[n \frac{z}{z-r} \right]$$

$$= n \left[-z \frac{d}{dz} \frac{z}{z-r} \right]$$

$$= n \left[-z \left(\frac{z-r-z}{(z-r)^2} \right) \right]$$

$$= n \frac{rz}{(z-r)^2}$$

applying z-transform again.

$$(z-r)^2$$

applying z-transform again.

$$= -z \frac{d}{dz} \left(\frac{rz}{(z-r)^3} \right)$$

$$= \frac{-z \{ (z-r)^2 r - 2rz(z-r) \}}{(z-r)^4}$$

$$= \frac{rz(r+z)}{(z-r)^3} \quad \text{Ans}$$

c) $n[n^2 u[n]]$

↑

as we have already solved this in part - a, so we will continue applying z-transform.

$$= n \frac{z(z+1)}{(z-1)^3}$$

$$= \frac{-z \{ (z-1)^3 (2z+1) - z(z+1) 3(z-1)^2 \}}{(z-1)^6}$$

$$= \frac{-z (2z^2 + z - 2z - 1 - 3z^2 - 3z)}{(z-1)^6}$$

$$= \frac{-z (-z^2 - 4z - 1)}{(z-1)^4} \quad \text{Ans}$$

d) $a^n [u[n] - u[n-m]]$

$$= a^n u[n] - a^n u[n-m]$$

$$= z \{ a^n u[n] \} - z \{ a^n u[n-m] \}$$

$$= \frac{z}{z-a} - z \left\{ a^m a^{n-m} u[n-m] \right\} \quad \left[\because a^n u[n] \xrightarrow{z} \frac{z}{z-a} \right]$$

$$= \frac{z}{z-a} - a^m z \left\{ a^{n-m} u[n-m] \right\}$$

$$= \frac{z}{z-a} \left[1 - a^m z^{-m} \right] \quad \text{Ans.}$$

Problem 4 [12 pts]

Consider a DT-LTI system given by the difference equation

$$y[n] - \frac{1}{4}y[n-2] = x[n-1]$$

- Use transform-domain techniques to determine the zero-state response $y_{zsr}[n]$ to input $x[n] = 3u[n-5]$.
- Use transform-domain techniques to determine the zero-input response $y_{zir}[n]$ given $y_{zir}[-2] = y_{zir}[-1] = 1$.

NOTE: Solve using the Z transform

a) we know that zero state response is the output response of a system solely due to input, assuming all initial conditions to be zero.
So,

$$y[-2] = y[-1] = 0$$

$$Y[z] - \frac{z^{-2}}{4} Y[z] = z \{ X[n-1] \}$$

$$Y[z] \left(1 - \frac{1}{4z^2} \right) = z^{-1} z \{ X[n] \}$$

$$x[n] = 3u[n-5]$$

$$x[n] = 3u[n-5] \quad \rightarrow \quad z^{-5} \rightarrow \quad z^{-4}$$

$$x[n] = 3u[n-5]$$

$$X[z] = 3z^{-5} \frac{z}{z-1} = \frac{3z^{-4}}{z-1}$$

$$Y[z] \left(\frac{4z^2-1}{4z^2} \right) = \frac{3z^{-4}}{z-1} \cdot z^{-1}$$

$$Y[z] \left(\frac{4z^2-1}{4z^2} \right) = \frac{3z^{-5}}{z-1}$$

$$Y[z] = \frac{4z^2}{(4z^2-1)} \cdot \frac{3z^{-5}}{z-1}$$

$$Y[z] = \frac{12z^{-3}}{(z-1)(4z^2-1)}$$

$$Y[z] = \frac{12}{z^3(z-1)(4z^2-1)}$$

$$Y[z] \cdot z^2 = \frac{12}{z(z-1)(2z-1)(z+1)}$$

Solve using partial fraction:-

$$\frac{12 \times (1/2) \times (1/2)}{z(z-1)(z-1/2)(z+1/2)} = \frac{A}{z} + \frac{B}{z-1} + \frac{C}{(z-1/2)} + \frac{D}{(z+1/2)} \quad \text{--- i}$$

$$= A(z-1)(z-1/2)(z+1/2) + B(z)(z-1/2)(z+1/2) + C(z)(z-1)(z+1/2) + D(z)(z-1)(z-1/2)$$

for $z=0$:

$$3 = A(-1)(-1/2)(1/2)$$

$$3 = A \left(\frac{1}{4} \right)$$

$$\boxed{A = 12}$$

- for $z = 1/2$:-

$$3 = 0 + 0 + C \left(\frac{1}{2} \right) \left(-\frac{1}{2} \right) (1) + 0$$

$$3 = \frac{1}{4} (-C)$$

$$\boxed{C = -12}$$

- for B :
 $z = 1$

$$3 = B \left(\frac{1}{2} \right) \left(\frac{3}{2} \right) + 0 + 0 + 0$$

$$B = \frac{4 \times 3}{3} = 4$$

- for D :

$$\text{let } z = -1/2:$$

$$3 = 0 + 0 + 0 + D \left(-\frac{1}{2} \right) \left(-\frac{3}{2} \right) (-1)$$

$$\boxed{D = -4}$$

Now, put the values in equation (i)

$$z^2 Y[z] = \frac{12}{z} + \frac{4B}{z-1} - \frac{12}{z-1/2} - \frac{4}{z+1/2}$$

Multiply by z on both sides:-

$$z^3 Y[z] = 12 + \frac{4z}{z-1} - \frac{12z}{z-1/2} - \frac{4z}{z+1/2}$$

$$z^3 Y[z] = 12 + \frac{4z}{z-1} - \frac{12z}{z-1/2} - \frac{4z}{z+1/2}$$

Now, inverse transform :-

$$z^{-1} \{ z^3 Y[z] \} = 12 \delta[n] + 4u[n] - 12 \left(\frac{1}{2}\right)^n u[n] - 4 \left(\frac{-1}{2}\right)^n u[n]$$

for $y[n]$:-

$$= 12 \delta[n-3] + 4u[n-3] - 12 \left(\frac{1}{2}\right)^{n-3} u[n-3] - 4 \left(\frac{-1}{2}\right)^{n-3} u[n-3]$$

Ans.

b) $y_z[-2] = y_z[-1] = 1$

$\therefore$ we know that for zero input response, we make the input zero, hence $x[n] = 0$

$$y[n] - \frac{1}{4} y[n-2] = 0$$

• Applying z-transform :-

$$z \{ y[n-2] \} = \sum_{n=0}^{\infty} y[n-2] z^{-n}$$

for $k = n-2$:

$$= \sum_{k=-2}^{\infty} y[k] z^{-k} z^{-2}$$

$$= z^{-2} \sum_{k=-2}^{\infty} y[k] z^{-k}$$

$$= z^{-2} [y[-2]z^2 + y[-1]z^1 + \sum_{k=0}^{\infty} y[k]z^{-k}]$$

$$z \{ y[n-2] \} = y[-2] + y[-1]z^{-1} + z^{-2} Y[z]$$

$$z \{ y[n-2] \} = 1 + z^{-1} + z^{-2} Y[z]$$

$$Y[z] - \frac{1}{4} - \frac{1}{4} z^{-1} - \frac{1}{4} z^{-2} Y[z] = 0$$

$$Y[z] \left(1 - \frac{1}{4} z^{-2} \right) = \frac{1+z^{-1}}{4}$$

$$Y[z] \left(\frac{4z^2 - 1}{4z^2} \right) = \frac{1+z^{-1}}{4}$$

$$Y[z] = \frac{z^2}{4z^2-1} (1+z^{-1}) = \frac{z^2+z}{4z^2-1}$$

$$\frac{Y[z]}{z} = \frac{(z+1)}{(2z-1)(2z+1)}$$

let:

$$\frac{z+1}{(2z+1)(2z-1)} = \frac{A}{(2z+1)} + \frac{B}{(2z-1)}$$

$$(z+1) = A(2z-1) + B(2z+1)$$

• for A:

$$\text{let } z = -1/2$$

$$-1/2 + 1 = A \left(2(-1/2) - 1 \right)$$

$$1/2 = A(-2)$$

$$\boxed{A = -1/4} \text{ Ans.}$$

• for B

$$\text{let } z = 1/2 :-$$

$$\frac{3}{2} = 0 + B(2)$$

$$\boxed{B = 3/4} \text{ Ans.}$$

Put the values in equation.

$$\frac{Y[z]}{z} = \frac{-1}{4} \cdot \frac{1}{(z+1)} + \frac{3}{4} \frac{1}{(z-1)}$$

$$Y[z] = \frac{3z}{8(z-1/2)} - \frac{z}{8(z+1/2)}$$

• Now, inverse z-transform:-

$$y[n] = \frac{3}{8} z^{-1} \left\{ \frac{z}{z-1/2} \right\} - \frac{1}{8} z^{-1} \left\{ \frac{z}{z+1/2} \right\}$$

$$y[n] = \frac{3}{8} \left(\frac{1}{2} \right)^n u[n] - \frac{1}{8} \left(-\frac{1}{2} \right)^n u[n]$$

So,

$$y[n] = \frac{3}{8} \left(\frac{1}{2} \right)^n - \frac{1}{8} \left(-\frac{1}{2} \right)^n, n \geq 0$$

Ans.

Problem 5 [12 pts]

Consider the transfer functions $H(z)$ of three different discrete-time LTI systems given below. For each system, determine all the poles and zeros and plot them in the z-domain along with the corresponding region of convergence (ROC). Using the pole-zero locations and ROC, determine whether each system is stable and invertible.

(a) $\frac{z(z-4)}{z^2-5z+6}$

(b) $\frac{z-4}{z^2-5z+6}$

(c) $\frac{(e^{-2}-2)z}{(z-e^{-2})(z-2)}$

a)

$$H(z) = \frac{z(z-4)}{z(z-2) \cdot z(z-3)}$$

$$= \frac{z(z-4)}{(z-2)(z-3)}$$

So,

as $z \neq 2, 3$

So,

as $z \neq 2, 3$

So,

Poles: $z=2, z=3$

Zeros:

$$z^2 - 4z = 0$$

$$z=0, z=4$$

- as ROC must not include any poles so

$$\text{ROC: } |z| > 3, 2 < |z| < 3, |z| < 2$$

- ROC: $|z| > 3$:

- Stable: NO

- Invertible: NO

ROC: $2 < |z| < 3$

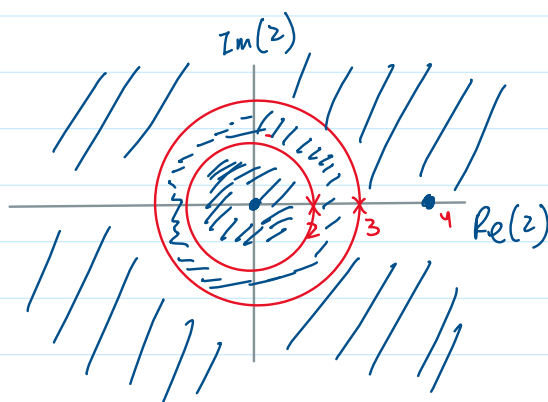
Stable: NO

Invertible: NO

ROC: $|z| < 2$

Stable: Yes

Invertible: NO



b) for zeros:-

$$z - 4 = 0$$

$$z = 4$$

Poles:-

$$z^2 - 5z + 6 = 0$$

$$\frac{z-4}{(z-2)(z-3)} = 0$$

as $z \neq 2 \neq 3$ so,
Poles: $z=2$, $z=3$

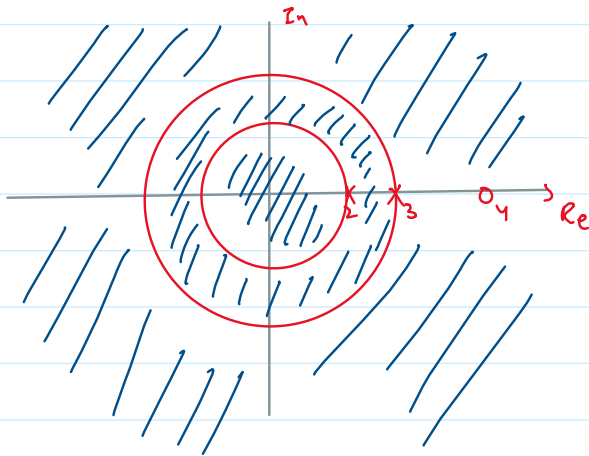
- So ROC: $|z| > 3$, $|z| < 2$, $2 < |z| < 3$

↓
Stable: NO
invertible: YES

↓
Stable: YES
invertible: NO

↓
Stable: NO
invertible: NO

- Invertible, NO pole-zero cancellation.



c) Poles:

$$(z - e^{-2})(z - 2) = 0$$

$$[z = e^{-2}, z = 2]$$

for zeros:

$$z = 0$$

- ROC, two sided system possible, $e^{-2} < |z| < 2$ = Stable: YES, invertible: NO
 $|z| < e^{-2}$ stable: NO, invertible: YES
 $|z| > 2$: stable: NO, invertible: NO
- stable, w/ circle included.
- NO cancellation.

